

PAUL M. CHICHURA

(570)557-1024 | pchichura@uchicago.edu | Department of A&A, 5640 S Ellis Ave, Chicago, IL 60637
pchichura.github.io | linkedin.com/in/pchichura | orcid.org/0000-0002-5397-9035

RESEARCH INTERESTS

AI-enabled telescope operations • time-variable millimeter-wave astronomy • astronomical survey design

APPOINTMENTS

2025-present **Postdoctoral Scholar**, NSF-Simons AI Institute for the Sky (SkAI), University of Chicago
Supervisor: Alex Drlica-Wagner

EDUCATION

2025 **Ph.D., Physics**, (*Nathan Sugarman Award, Sachs Fellowship*), University of Chicago
Advisor: Thomas Crawford
2022 **M.S., Physics**, University of Chicago
2018 **M.S., Physics and Astronomy**, University of Pennsylvania
2018 **B.A., Physics**, (*Summa Cum Laude, dept. honors, Phi Beta Kappa*), University of Pennsylvania
Advisor: James Aguirre
2014-2015 Schreyer Honors College, The Pennsylvania State University

RESEARCH EXPERIENCE

2025-present **NSF-Simons AI Institute for the Sky (SkAI)**

- Developing AI agents for autonomous telescope scheduling, trained on 6 years of archival DES data, with deployment on the NSF's 4-meter Blanco Telescope control system

2020-2025 **South Pole Telescope (SPT) and Event Horizon Telescope (EHT)**

- Designed and deployed ML control models for SPT using real-time sensor data streams, reducing errors pointing at the sky by 33%
- Built analysis pipeline for asteroid detection in CMB survey data, enabling the first targeted mm-wave detections and publishing new standard methods
- Analyzed SPT time-series observations of Sgr A* and created a real-time calibration pipeline to reduce systematic errors across ~16,000 detectors
- Designed and led new Galactic Center survey observations with SPT

2017-2019 **Hydrogen Epoch of Reionization Array (HERA)**

- Developed power-spectrum data-analysis pipelines for HERA commissioning array data

HONORS AND AWARDS

2024 Nathan Sugarman Award for Excellence in Graduate Student Research, University of Chicago
2021 Sachs Fellowship, University of Chicago
2021 Chambliss Astro. Achievement Student Award Honorable Mention, American Astronomical Society
2018 Phi Beta Kappa Honor Society, University of Pennsylvania
2015 The President's Freshman Award, The Pennsylvania State University

PROFESSIONAL SERVICE AND LEADERSHIP

2026-present Co-chair, Local Organizing Committee, Open SkAI 2026 Conference, SkAI
2024-2025 Analysis Coordinator, Sources/Transients Working Group, SPT Collaboration, University of Chicago
2022-2025 Program Lead, First Discoveries outreach initiative, SPT Collaboration, University of Chicago
2021-2024 Coordinator, Climate Surveys, Chicago SPT Collaboration, University of Chicago
2023 Referee, Progress of Theoretical and Experimental Physics (PTEP) Journal

RESEARCH MENTORSHIP

Graduate Students (1):

2023-present Bhaskar Mondal (*project advisor, non-voting committee member*) – Ph.D. student; Aerospace Engineering; University of Illinois Urbana-Champaign. Detecting and thermal modeling asteroids in SPT data.

Undergraduate Students (3):

2023 Mariah Jones (*co-supervised with Thomas Cranford*) – Astrophysics major; Vassar College; summer REU at University of Chicago. Thermal modeling of asteroids with microwave data.

2022 Marvin Lopez Acevedo (*co-supervised with Alexandra Rablin*) – Physics, Geosciences major; Hamilton College; summer REU at University of Chicago. Built monitor webpages for SPT-SZ cryostat.

2021 Nahuel Ossa-Jaen (*co-supervised with Thomas Cranford*) – Physics, Computer Science major; University of Redlands; summer REU at University of Chicago. Detecting asteroids in SPT CMB data.

TEACHING

2025 College Teaching Certificate, University of Chicago

2024 Guest Lecturer, University of Chicago, *ASTR 12620: The Big Bang*

2019-2020 Teaching Assistant, University of Chicago
PHSC 116: Phys. for Future Presidents, PHSC 117: Phys. for Future Presidents, PHYS 123: General Phys. III

2018-2019 Teaching Assistant, University of Pennsylvania
PHYS 101: General Physics: Mechanics, PHYS 102: General Physics Electricity and Magnetism

OUTREACH AND PUBLIC EDUCATION

2020-2025 Program Lead & Classroom Volunteer, First Discoveries¹, University of Chicago

- Designed, published, taught early-childhood science lessons at John Fiske Elementary School
- Co-designed, co-led 2 professional development trainings for 15 teachers each, 2023 and 2024

2021-2023 Discussion Organizer, Chicago-SPT Collaboration, University of Chicago

- Initiated, conducted 3 climate surveys of the Chicago-SPT group

2022 Volunteer, South Side Science Festival, UChicago

INVITED TALKS

2026 **Invited Conference Talk**, “*Self-Driving Telescopes: Intelligent Scheduling for Astronomical Surveys*”
CosmicAI’s Cosmic Horizons 2026, National Radio Astronomy Observatory

2024 **Award Talk**, “*Novel Science with the South Pole Telescope: Asteroids and Black Holes*”
Nathan Sugarman Awards, University of Chicago

2022 **Seminar Talk**, “*Solar System Science with CMB-S4*”
Center for AstroPhysical Surveys (CAPS) Seminar, University of Illinois Urbana-Champaign

CONTRIBUTED TALKS AND POSTERS

2026 **Contributed Talk**, “*Telescope Control with Machine Learning: Pointing Corrections and Observation Scheduling*”
USNC National Radio Science Meeting, University of Colorado Boulder

2025 **Contributed Talk**, “*Pointing the South Pole Telescope with Machine Learning*”
USNC National Radio Science Meeting, University of Colorado Boulder

2024 **Poster**, “*Pointing Model Predictions with Machine Learning for the South Pole Telescope*”
AI+Science Summer School, University of Chicago

2023 **Poster**, “*Pointing Model Predictions for the South Pole Telescope*”
AI+Science Summer School, University of Chicago

2023 **Contributed Talk**, “*Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope*”
Transient and Variable Universe 2023, University of Illinois Urbana-Champaign

¹ <https://pole.uchicago.edu/public/First%20Discoveries.html>

- 2022 **Talk**, “*Asteroids and CMB-S*,”
CMB-S4 Astrophysics Workshop, University of Illinois Urbana-Champaign (remote)
- 2021 **Poster**, “*Measuring the Millimeter-Wavelength Flux of Asteroids with the South Pole Telescope*”
238th Meeting of the American Astronomical Society, remote
- 2018 **Poster**, “*Polarized Power Spectra from HERA-19 Commissioning Data: Effect of Calibration Techniques*”
231st Meeting of the American Astronomical Society, Washington, D.C.

SELECTED PUBLICATIONS

- [1] Wan, Y., J. Vieira, **P. Chichura**, *et al.* 2026. “Detection of Millimeter-Wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey.” *ApJ* 997, no. 2 (Feb.): 133. [arXiv:2509.08962](#).
▪ Designed and led the observation survey that resulted in data analyzed in this paper
- [2] **Chichura, P.**, *et al.* 2025. “Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning.” *Journal of Astronomical Instrumentation* 14 (January): 2550001. [arXiv:2412.15167](#).
▪ Led analysis, development, and deployment of models; a substantial part of Ph.D. thesis
- [3] **Chichura, P.**, *et al.* 2022. “Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope.” *ApJ* 936, no. 2 (September): 173. [arXiv:2202.01406](#).
▪ Led analysis and development of novel techniques; a substantial part of Ph.D. thesis
- [4] Kohn, S., *et al.* 2019. “The HERA-19 Commissioning Array: Direction-dependent Effects.” *ApJ* 882, no. 1 (September): 58. [arXiv:1802.04151](#).
▪ Results from undergraduate honors thesis research

ALL PUBLICATIONS (33 refereed publications, 951 total citations, h-index: 16)

- [1] Archipley, M., *et al.* 2026. “Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs.” *A&A* 706 (January): A17. [arXiv:2506.00298](#).
- [2] Bernshteyn, V., *et al.* 2026. “Ring Asymmetry and Spin in M87*.” *ApJ* 1000, no. 2 (April): 231. [arXiv:2601.00394](#).
- [3] Camphuis, E., *et al.* 2026. “SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field.” *PhRvD* 113, no. 8 (April): 083504. [arXiv:2506.20707](#).
- [4] Chaubal, P., *et al.* 2026. “SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power.” *arXiv e-prints* (January): [arXiv:2601.20551](#).
- [5] de Haan, T., *et al.* 2026. “Characterization of the Polarization Beam Response of SPT-3G Using Point Sources.” *arXiv e-prints* (February): [arXiv:2602.06334](#).
- [6] Georgiev, B., *et al.* 2026. “Locating the missing large-scale emission in the jet of M87* with short EHT baselines.” *A&A* 709 (April): A26. [arXiv:2601.13356](#).
- [7] Maniyar, A. S., *et al.* 2026. “SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys.” *arXiv e-prints* (February): [arXiv:2602.11279](#).
- [8] Qu, F., *et al.* 2026. “Unified and consistent structure growth measurements from joint ACT, SPT and Planck CMB lensing.” *PhRvL* 136, no. 2 (January): 021001. [arXiv:2504.20038](#).
- [9] Quan, W., *et al.* 2026. “SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field.” *arXiv e-prints* (March): [arXiv:2603.20163](#).
- [10] Raghunathan, S., *et al.* 2026. “Measurement of the Full Shape of the Thermal Sunyaev-Zel’dovich Power Spectrum from the South Pole Telescope and HerschelSPIRE Observations.” *ApJL* 1002, no. 1 (May): L22. [arXiv:2602.10107](#).
- [11] Saurabh, H. *et al.* 2026. “Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations.” *A&A* 706 (January): A27. [arXiv:2512.08970](#).
- [12] Tandoi, C., *et al.* 2026. “Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data.” *arXiv e-prints* (May): [arXiv:2605.01022](#).
- [13] Wan, Y., J. Vieira, **P. Chichura**, *et al.* 2026. “Detection of Millimeter-Wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey.” *ApJ* 997, no. 2 (Feb.): 133. [arXiv:2509.08962](#).
- [14] **Chichura, P.**, *et al.* 2025. “Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning.” *Journal of Astronomical Instrumentation* 14 (January): 2550001. [arXiv:2412.15167](#).
- [15] Coerver, A., *et al.* 2025. “Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G.” *ApJ* 982, no. 1 (March): 15. [arXiv:2407.20579](#).

- [16] Event Horizon Telescope Collaboration. 2025. “Horizon-scale variability of M87* from 2017–2021 EHT observations.” *A&A* 704 (December): A91. [arXiv:2509.24593](#).
- [17] Foster, A., *et al.* 2025. “Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites.” *The Open Journal of Astrophysics* 8 (May): 51. [arXiv:2411.03374](#).
- [18] Ge, F., *et al.* 2025. “Cosmology from CMB lensing and delensed EE power spectra using 2019–2020 SPT-3G polarization data.” *PhRvD* 111, no. 8 (April): 083534. [arXiv:2411.06000](#).
- [19] Khalife, A., *et al.* 2025. “SPT-3G D1: Axion Early Dark Energy with CMB experiments and DESI.” *arXiv e-prints* (July): [arXiv:2507.23355](#).
- [20] Kornoelje, K., *et al.* 2025. “The SPT-Deep Cluster Catalog: Sunyaev-Zel’dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees.” *arXiv e-prints* (March): [arXiv:2503.17271](#).
- [21] Vitrier, A., *et al.* 2025. “Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky.” *arXiv e-prints* (October): [arXiv:2510.24669](#).
- [22] Zebrowski, J., *et al.* 2025. “Constraints on Inflationary Gravitational Waves with Two Years of SPT-3G Data.” *PhRvD* 112, no. 12 (December): 123520. [arXiv:2505.02827](#).
- [23] Ansarinejad, B., *et al.* 2024. “Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing.” *JCAP* 2024, no. 7 (July): 024. [arXiv:2404.02153](#).
- [24] Prabhu, K., *et al.* 2024. “Testing the Λ CDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G.” *ApJ* 973, no. 1 (September): 4. [arXiv:2403.17925](#).
- [25] Raghunathan, S., *et al.* 2024. “First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel’dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations.” *PhRvL* 133, no. 12 (September): 121004. [arXiv:2403.02337](#).
- [26] Tandoi, C., *et al.* 2024. “Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G.” *ApJ* 972, no. 1 (September): 6. [arXiv:2401.13525](#).
- [27] Balkenhol, L., *et al.* 2023. “Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset.” *PhRvD* 108, no. 2 (July): 023510. [arXiv:2212.05642](#).
- [28] Pan, Z., *et al.* 2023. “Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data.” *PhRvD* 108, no. 12 (December): 122005. [arXiv:2308.11608](#).
- [29] Schiappucci, E., *et al.* 2023. “Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel’dovich effect with SPT-3G and DES.” *PhRvD* 107, no. 4 (February): 042004. [arXiv:2207.11937](#).
- [30] **Chichura, P.**, *et al.* 2022. “Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope.” *ApJ* 936, no. 2 (September): 173. [arXiv:2202.01406](#).
- [31] Ferguson, K., *et al.* 2022. “Searching for axionlike time-dependent cosmic birefringence with data from SPT-3G.” *PhRvD* 106, no. 4 (August): 042011. [arXiv:2203.16567](#).
- [32] Ghosh, A., *et al.* 2020. “Foreground modelling via Gaussian process regression: an application to HERA data.” *MNRAS* 495, no. 3 (January): 2813–2826. [arXiv:2004.06041](#).
- [33] Kohn, S., *et al.* 2019. “The HERA-19 Commissioning Array: Direction-dependent Effects.” *ApJ* 882, no. 1 (September): 58. [arXiv:1802.04151](#).